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virtualis fit realis

Declaration

I hereby declare that this thesis is my own work and that all sources and citations have been properly acknowledged. No part of this thesis has been submitted for any other degree or professional qualification at this or any other institution.

Abstract

The abstract should briefly describe the context of the research, the central research question, the methodology used, the most important results, and the main conclusion. Keep it concise and informative, aiming to give readers a clear overview of the entire work.

Acknowledgements

I would like to thank [Name] for [their] guidance and support throughout the course of this thesis. I also wish to express my gratitude to [institution/lab] for providing the resources necessary to carry out this research, and to my family and friends for their encouragement.

Abbreviations

SoE Sense of Embodiment

SoO Sense of Ownership (also known as sense of body ownership, SoBO)

SoA Sense of Agency

SoL Sense of Self-Location

VR Virtual Reality

XR Extended Reality

IK Inverse Kinematics

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Chapter 1

Introduction

1.1 Motivation

Indoor rowing is a physically demanding and technically structured form of exercise. Unlike many stationary fitness activities, rowing involves a coordinated movement of legs, trunk, and arms, expressed on the ergometer through seat, handle, and stroke rhythm. On a rowing ergometer, this movement is translated into sliding-seat and handle motion on a stationary training device. Ergometer rowing can reproduce important aspects of rowing training, but it is not identical to rowing on water, especially where boat motion, oar handling, and the interaction between oar and water shape the stroke (Lamb, 1989; Rauter et al., 2013; Soper & Hume, 2004)¹

Virtual reality offers an opportunity to reintroduce some of the experiential context that the ergometer removes. A virtual rowing system can place the user in a boat, on a river, or in a training environment that visually resembles the sport more closely than a conventional ergometer display. Prior

¹*Source check:* The Motivation opens from rowing and ergometer mechanics, not from VR theory. Soper and Hume support treating ergometer rowing as a normal rowing training and testing context while also warning that common ergometers reproduce only parts of on-water body action, especially less well for trunk and upper-limb patterns. Lamb gives the direct ergometer-versus-on-water kinematic comparison: leg and trunk movement are broadly similar in many drive-phase variables, but upper arm and forearm movement differ, especially around catch and finish, partly because on-water rowing includes oar handling such as lifting and feathering that is absent on the ergometer. Rauter et al. support the broader sport-action premise that rowing involves coordinated body movement and continuous haptic interaction among rower, boat, oar, and water, while ordinary indoor ergometers simplify this interaction through cable-driven resistance and do not train oar handling or oar height in water. Together, these sources justify the thesis wording that ergometer rowing is related to, but not identical with, on-water rowing. They do not support claims about VR embodiment outcomes or user preference.

rowing and VR sport systems have shown how stationary exercise can be connected to virtual environments, avatars, social or competitive context, and visually represented movement through space (Arndt et al., 2018; Murray et al., 2016; Neumann et al., 2018). In such systems, visual motion cues such as optic flow may also contribute to an impression of self-motion, even when the user’s physical body remains in one place (Niehorster, 2021; Palmisano et al., 2015)²

However, this opportunity also introduces a design problem. The ergometer makes rowing available indoors by replacing the boat and oars with a stationary machine. Virtual reality can visually restore the boat, oars, water, and movement through an environment, but it cannot remove the fact that the user is still physically rowing on an ergometer. The system must therefore decide how the real movement should become a virtual rowing action.³

²*Source check:* This citation cluster establishes VR as a way to restore missing rowing context, while avoiding the stronger claim that prior work studies this thesis’ exact mapping comparison. Murray et al. provide a concrete prior VR rowing example in which participants row on an ergometer while seeing a virtual river, avatar timing, and companion/social conditions. Arndt et al. provide an HMD rowing prototype in which sensor-derived ergometer movement drives a virtual rowing scene with lake/scull feedback, while the real haptic feedback still comes from the physical rowing machine. Neumann et al. provide the broader interactive VR sport frame and include rowing examples where exertion on an ergometer can be represented as virtual oar movement and movement through water or scenery. Niehorster supports the narrow optic-flow wording by defining optic flow as global visual motion that carries self-motion information. Palmisano et al. support the cautious self-motion/vection background by discussing subjective self-motion in stationary observers. The self-motion sentence remains phrased as “may contribute” because this thesis does not measure vection and should not claim that optic flow automatically produces embodiment, agency, ownership, or preference.

³*Source check:* This paragraph is the first explicit thesis-synthesis point. The rowing sources establish the first premise: ordinary ergometer rowing is a legitimate indoor rowing form, but it replaces or alters the boat-oar-water interaction. Soper and Hume support common ergometer use and biomechanical caveats; Lamb directly compares ergometer and on-water rowing and identifies upper-limb/oar-related differences; Rauter et al. emphasize that rowing involves continuous rower-boat-oar-water interaction while ordinary ergometers use cable-driven resistance and do not train oar handling or oar height. The VR sources establish the second premise: virtual rowing systems can visually present rowing environments and couple ergometer effort to virtual action. Murray et al. use an ergometer with a virtual river/avatar setup, Arndt et al. use an HMD rowing prototype with a virtual lake/scull driven by ergometer sensor data, and Neumann et al. describe interactive VR sport in which rowing handle pulls can become virtual oar movements and exertion can drive faster movement through water and scenery. The thesis inference is that VR rowing must choose how the real ergometer movement becomes a virtual rowing action. This should not be overread as a claim that VR restores full on-water haptics or that prior work has already studied the exact two-mode comparison.

This decision matters because indoor rowing in VR is not simply a question of adding a more attractive background to exercise. The user performs a real, repetitive, forceful movement while seeing a mediated version of that movement in a virtual environment. The virtual scene can make the activity visually resemble rowing on water, but the physical interface remains the ergometer. As a result, the system must balance two different forms of fidelity: fidelity to the user’s actual sensorimotor interaction with the machine, and fidelity to the recognizable visual and sport-specific form of rowing.⁴

This thesis investigates that balance. It examines VR indoor rowing as a case where a physical exercise device stands in for a more complex sport action. The central question is whether the virtual representation should preserve the user’s actual interaction with the ergometer as directly as possible, or transform that interaction into the boat- and oar-based action associated with rowing on water.⁵

⁴*Source check:* This paragraph reframes VR rowing as an action-mapping problem rather than background decoration. Neumann et al. support this by defining interactive VR sport through a simulated sport environment and athlete interactivity, and by giving the rowing example where ergometer handle pulls become virtual oar movements and exertion changes virtual speed through water and scenery. Murray et al. and Arndt et al. provide concrete VR rowing systems in which real ergometer movement is mediated as avatar, boat, scull, river, lake, or scenery movement. Rauter et al. and Lamb explain why this is not a trivial one-to-one mapping: rowing is a rower-boat-oar-water interaction, ordinary ergometers replace important oar-water mechanics, and ergometer/on-water kinematics differ especially around upper-limb and oar-related parts of the stroke. The thesis inference is the two-fidelity tension: fidelity to the user’s actual sensorimotor interaction with the machine versus fidelity to the recognizable visual and sport-specific form of on-water rowing. The sources support the premises for this tension, but not any conclusion that one form of fidelity is superior.

⁵*Source check:* This closing Motivation paragraph states the thesis’ own research focus. External sources should only support the premises, not be presented as if they posed the exact central question. Rauter et al. support the idea that rowing is a complex sport action involving coordinated body movement and rower-boat-oar-water interaction, while ordinary ergometers omit oar handling and oar height in water. Lamb supports the kinematic mismatch that makes preserving actual ergometer movement and transforming it into boat/oar action meaningfully different choices. Neumann et al. support the interactive VR sport premise that physical exertion on an interface can become virtual sport action, including the example of rowing handle pulls represented as virtual oar movements. The thesis-authored question is whether the virtual representation should stay close to the user’s real ergometer interaction or transform that interaction into the recognizable boat-and-oar action of on-water rowing. This paragraph should stay comparative and open-ended, without implying that one mapping is already known to be superior.

1.2 The Core Design Tension

A rowing ergometer and a rowing boat do not provide the same movement interface. On an ergometer, the user sits on a sliding seat and pulls a handle through constrained handle movement. In a rowing boat, the rower interacts with oars that rotate around oarlocks, blades that interact with water, and a boat that moves relative to the environment. Even when the training movement is related, the physical apparatus and spatial logic are different (Lamb, 1989; Rauter et al., 2013; Soper & Hume, 2004)⁶

A VR rowing system can respond to this difference in at least two plausible ways. This thesis refers to these strategies as the *erg-centered* and *boat-centered* modes. In the erg-centered mode, the virtual body, handle, and rowing apparatus are aligned primarily with the user’s real machine and movement. The visible hands and handle can remain close to where the user feels them, and the avatar can be organized around the physical ergometer interaction rather than around virtual oar geometry. This preserves the physical logic of the user’s current interaction and prioritizes visuomotor correspondence between the performed rowing movement and the movement shown in VR (Kilteni et al., 2012)⁷

⁶*Source check:* This paragraph grounds the design tension in rowing mechanics rather than in VR theory. Lamb directly compares ergometer and on-water rowing and supports the nuanced premise that the movements are related but not identical: leg and trunk patterns are broadly similar in many drive-phase variables, while upper arm and forearm patterns differ, especially around catch and finish where on-water rowing involves oar-water mechanics such as lifting and feathering that are absent from ergometer rowing. Soper and Hume support the status of the ergometer as a normal rowing training/testing device while also noting that common ergometers reproduce only parts of on-water body action and do not reproduce trunk and upper-limb patterns particularly well because of the central pulley system. Rauter et al. support the boat-side interface: rowing involves continuous rower-boat-oar-water interaction, ordinary ergometers replace this with cable-driven resistance, and realistic rowing includes oar handling, blade-water interaction, oar angles, and boat propulsion. The thesis inference is that these source-backed differences amount to different movement interfaces and spatial action logics. These sources do not support VR embodiment outcomes or any claim that the ergometer is invalid as training equipment.

⁷*Source check:* This paragraph defines the thesis’ erg-centered mode rather than importing a named category from the literature. The term is thesis shorthand for a representation strategy that keeps the virtual body, handle, apparatus, and avatar movement organized around the user’s real ergometer interaction. Kilteni et al. are cited only for the theoretical relevance of visuomotor correspondence: their sense-of-embodiment framework links agency to comparisons between predicted and actual sensory consequences and to synchronous visuomotor correlations under active movement. This source does not define the erg-centered mode, does not study rowing, and does not prove that this mode will produce stronger embodiment in the planned study. The implementation-specific details

In the boat-centered mode, the system instead prioritizes the visual and sport-specific logic of rowing on water. The user may appear to sit in a boat, move through a river, and row with oars. This can make the virtual scene more recognizable as rowing, but it also means that the real ergometer handle movement must be interpreted as a virtual boat-and-oar action. The resulting virtual oar movement may therefore remain responsive to the user’s stroke while no longer coinciding fully with the felt position and movement of the physical handle (Harris et al., 2019; Lamb, 1989; Neumann et al., 2018; Rauter et al., 2013)⁸

The comparison is therefore not between a real and an artificial representation of rowing. Both modes are representations of an indoor rowing action under different constraints. The erg-centered mode preserves the physical logic of the user’s actual device and movement, but can still place that indoor action inside a mediated rowing environment. The boat-centered mode preserves the recognizable boat-and-oar logic of the sport, but must transform the real ergometer interaction into a virtual oar action. There is no neutral third option in which the indoor ergometer simply becomes on-water rowing without remainder (Harris et al., 2019; Lamb, 1989; Neumann et al., 2018; Rauter et al., 2013)⁹

of how handle, seat, and body state are tracked or inferred belong in Chapter 3.

⁸*Source check:* This paragraph defines the thesis’ boat-centered mode rather than importing a named category from the literature. Neumann et al. support the general VR sport mapping premise: physical effort on an exercise interface can be translated into virtual sport performance, and the review gives rowing examples where ergometer handle pulls can be represented as virtual oar movements and greater exertion can drive faster virtual movement through water and scenery. Rauter et al. support the rowing-specific boat-and-oar side of the paragraph: realistic rowing involves rower, boat, oar, blade, oarlock, and water interaction, including oar handling and blade-water mechanics that ordinary ergometers do not reproduce. Lamb supports the mechanical mismatch by showing that ergometer and on-water rowing are related but differ in upper-limb and oar-related movement around catch and finish. Harris is used as a general VR action caution: visually convincing VR does not automatically preserve the sensory and haptic conditions of real-world action. Together, these sources justify the paragraph’s thesis inference that a boat-centered representation can be visually recognizable as rowing while still transforming the felt ergometer handle movement into a virtual oar action. They do not prove that the boat-centered mode will be preferred or embodied more strongly.

⁹*Source check:* This paragraph is a conceptual guardrail. It prevents the comparison from being read as real rowing versus fake rowing or correct rowing versus wrong rowing. Lamb and Rauter support the mechanical premise that ergometer rowing and on-water rowing are related but not interchangeable: Lamb documents upper-limb and oar-related differences between ergometer and on-water rowing, while Rauter emphasizes the rower-boat-oar-water interaction that ordinary ergometers simplify. Neumann supports the VR sport premise that physical effort on an interface can be translated into virtual sport action, including rowing examples where handle pulls become virtual oar movements.

This makes VR indoor rowing a useful case for studying a broader XR design problem. Many XR sport and exercise systems use stationary or simplified physical equipment to represent more complex activities. The system can either make the virtual action correspond closely to the user’s real movement, or it can transform the movement to better match the represented activity. Both choices solve one part of the problem while introducing another.

In this thesis, this design problem is framed as a trade-off between visuomotor congruence and realistic sport depiction. Visuomotor congruence refers to the correspondence between the user’s performed movement and the movement they see in the virtual environment. Realistic sport depiction refers to the extent to which the virtual scene and action resemble the sport being represented. The two can support each other, but in indoor rowing they can also come apart.

1.3 Embodiment and Experiential Evaluation

The trade-off between ergometer-congruent and boat-centric representation is especially relevant for virtual embodiment. Virtual embodiment concerns the extent to which users experience a virtual body or body part as related to their own body. Commonly discussed components include the sense of agency over the virtual body’s actions, the sense of body ownership, and self-location in relation to the virtual body or space.

The ergometer-congruent condition is closely related to principles from virtual hand and body ownership research. In such paradigms, embodiment can be influenced by whether seen and felt body positions, movements, or touches are spatially and temporally aligned. In the rowing case, the real and virtual hands or handle can potentially be placed in close correspondence: the user feels the physical handle and sees the virtual hands or handle move in the same way.

However, rowing also differs from classic hand-ownership paradigms. It is an active, rhythmic, tool-mediated, whole-body exercise rather than a passive illusion involving a single limb. The user does not merely observe a virtual hand; they exert force, coordinate multiple body segments, and interact with

Harris supports the caution that visually convincing VR action does not automatically reproduce the sensory and haptic conditions of real-world action. The thesis inference is that both modes are compromises: the erg-centered mode preserves the user’s actual physical interaction but still mediates it in VR, while the boat-centered mode preserves the recognizable boat-and-oar form of the sport but must transform the physical ergometer interaction. The sources support the premises for this framing, but the statement that there is no neutral third option is thesis synthesis.

a physical machine. This makes indoor rowing an interesting extension of embodiment questions into a sport and exercise context.

The boat-centric condition may be less directly congruent with the user’s felt hand and handle movement, but it may better express what users understand as rowing. A virtual boat, oars, water, and forward movement can contribute to perceived sport realism and plausibility. For a rower or rowing-interested user, this representation may feel more meaningful or more appropriate to the activity, even if the mapping between real and virtual movement is less direct.

For this reason, the thesis does not treat embodiment as the only relevant outcome. The primary focus remains on embodiment-related experience, particularly agency, ownership, self-location, and overall sense of embodiment. However, the study also considers perceived rowing realism and user preference as design-facing measures. A representation can be sensorimotor-congruent but still be experienced as a poor depiction of rowing; conversely, a representation can be less congruent but preferred because it better captures the intended sport scenario.

Preference is therefore treated as an exploratory complement rather than as the main theoretical construct. It is included because practical XR sport systems are not evaluated only by whether they produce embodiment, but also by whether users find the representation convincing, appropriate, and desirable for the activity. At the same time, preference must be interpreted cautiously, because it may depend on visual polish, prior rowing experience, individual expectations, and details of the implementation.

1.4 Research Gap

Research on virtual embodiment has shown that multisensory congruence, visuomotor synchrony, and avatar representation can influence how users experience virtual bodies and actions.

A separate body of work has explored virtual reality and interactive systems for sport, exercise, and rowing. These systems often aim to support training, motivation, feedback, or presence by augmenting stationary exercise with virtual environments, avatars, or simulated sport contexts.

Existing rowing-related VR and simulation work already contains many of the ingredients relevant to this thesis. Some systems use ergometer like vehicles as virtual representations. Others use 3D models of rowing machines or rowing machine inspired vehicles to represent the User. Some systems use inverse kinematics or avatar animation to represent the user’s rowing movement. Commercial and consumer-facing applications also illustrate differ-

ent representational choices, ranging from travelling through virtual environments on an indoor machine to more boat- or oar-centered rowing depictions.

However, the specific comparison addressed in this thesis appears to be missing from the reviewed literature. Existing studies tend to compare VR against non-VR rowing, evaluate feedback or visualization techniques, manipulate avatar properties, adjust simulator parameters, or study motivational and social factors. They generally do not isolate the physical-to-virtual rowing mapping itself as the central experimental variable.

More specifically, the reviewed material does not appear to contain a controlled study that holds the physical rowing setup constant while comparing an ergometer-based mapping against a boat-based mapping as the main independent variable. Such a comparison would ask whether the avatar and rowing apparatus should be solved primarily in the coordinate frame of the real ergometer, or transformed into the coordinate frame of a virtual boat and oar system.

This gap is important because the mapping choice is not only a technical implementation detail. It changes the relation between the user’s body, the physical machine, and the virtual action they see. As a result, it may influence agency, body ownership, self-location, perceived realism, and preference. If the mapping is not isolated experimentally, it remains unclear whether a more sport-realistic rowing depiction supports the experience more strongly than a more sensorimotor-congruent ergometer depiction, or whether the opposite is the case.

The thesis addresses this gap by focusing on the mapping choice itself. It compares an ergometer-congruent representation and a boat-centric representation within the same general VR rowing context. The aim is not to evaluate every aspect of VR rowing, but to examine how this specific design choice shapes embodied and experiential responses.

1.5 Research Objectives and Questions

The objective of this thesis is to investigate how different representation strategies in VR indoor rowing influence embodiment-related and experiential outcomes. Specifically, the thesis compares an ergometer-congruent representation with a boat-centric representation. The comparison is guided by the assumption that both strategies can be justified by different design goals: the ergometer-congruent mode may support sensorimotor alignment and agency, while the boat-centric mode may support perceived rowing realism and the impression of rowing on water.

The first objective is to design and implement a VR rowing prototype that

can represent the same physical ergometer activity in two different ways. In the ergometer-congruent mode, the system emphasizes direct correspondence between the user’s real movement and the virtual body or handle. In the boat-centric mode, the system emphasizes a rowing scene with boat and oar depiction, even if this requires transforming the user’s handle movement into a more sport-like visual movement.

The second objective is to use this prototype as a research instrument. The system is not only an application prototype, but also a means of examining how physical-to-virtual movement mapping affects user experience. This requires the two modes to be comparable in as many respects as possible, so that the central difference remains the relation between the real ergometer movement and the virtual rowing depiction.

The third objective is to evaluate the two representation modes with respect to embodiment-related constructs. These include sense of agency, body ownership, self-location, and overall sense of embodiment. In addition, the study considers perceived rowing realism, comfort, and preference as exploratory design-facing outcomes.

The thesis is guided by the following main research question:

How does the trade-off between visuomotor congruence and realistic sport depiction affect embodiment, perceived rowing realism, and user preference in VR indoor rowing?

This main question is divided into the following subquestions:

1. How can an ergometer-congruent and a boat-centric rowing representation be implemented while keeping the physical task and virtual rowing context broadly comparable?
2. How do the two representation modes differ in embodiment-related experience, particularly sense of agency, body ownership, self-location, and overall sense of embodiment?
3. How do the two representation modes differ in perceived rowing realism, comfort, and user preference?
4. What do the results suggest for the design of XR sport and exercise systems in which physical equipment and virtual sport actions do not fully coincide?

These questions are intentionally framed as comparative and exploratory rather than as claims of expected superiority. It is plausible that the ergometer-congruent mode supports stronger agency because the visible movement more

closely matches the user’s physical action. It is also plausible that the boat-centric mode supports a stronger sense of rowing realism because it better represents the visual and cultural form of the sport. The study therefore examines how these dimensions interact rather than assuming that one form of fidelity is always preferable.

1.6 Contributions

This thesis makes three main contributions.

First, it presents the design and implementation of a VR indoor rowing prototype that supports two distinct representation strategies for the same physical rowing activity. The prototype distinguishes between an ergometer-congruent mode and a boat-centric mode. This distinction makes it possible to examine a design trade-off that is often implicit in XR sport systems: whether virtual representation should prioritize the user’s actual input apparatus and body movement, or the visual form of the sport being simulated.

Second, the thesis contributes an embodiment-oriented comparison of these two representation strategies. Rather than treating immersion, realism, or motivation as broad and undifferentiated goals, the thesis focuses on how movement mapping may affect specific components of embodied experience. In particular, it considers sense of agency, body ownership, self-location, and overall sense of embodiment as relevant constructs for evaluating rowing representation in VR.

Third, the thesis provides design implications for researchers and developers of XR sport and exercise systems in which physical equipment and virtual sport depiction do not fully coincide. The rowing ergometer is a clear example of this situation, but the underlying problem also applies to other stationary training systems where real input devices are used to represent more complex sport actions. The thesis therefore contributes to a broader discussion about how embodied XR systems should balance sensorimotor congruence, visual plausibility, sport-specific realism, and user preference.

The intended contribution is not a general claim that one representation strategy is universally superior. Instead, the thesis aims to clarify what is gained and what may be lost when a VR sport system chooses between direct physical congruence and realistic sport depiction. Both strategies solve part of the design problem while introducing another. A congruent representation may feel controllable and bodily grounded, while a sport-realistic representation may feel more meaningful or recognizable as rowing. Understanding this trade-off can help future systems make more deliberate representation choices.

In practical terms, the thesis also documents the technical and methodological considerations involved in building such a comparison. These include alignment between physical and virtual rowing spaces, avatar motion mapping, control of visual confounds, and the selection of measures for evaluating embodiment and experience. These details are part of the contribution because the comparison depends not only on the final visual result, but on the system’s ability to isolate the relevant design difference.

1.7 Thesis Structure

The remainder of this thesis is structured as follows.

Chapter 2 introduces the theoretical and empirical background for the work. It discusses virtual embodiment, sense of agency, body ownership, self-location, visuomotor congruence, and related concepts such as presence and vection. It also reviews relevant work on VR exercise, sport simulation, and rowing-related systems. The chapter establishes the conceptual basis for treating movement mapping as a central design factor in VR rowing.

Chapter 3 describes the VR rowing prototype developed for this thesis. It presents the hardware and software setup, the alignment and tracking approach, and the two representation modes. The ergometer-congruent mode and boat-centric mode are described in terms of their mapping strategies, avatar motion, and intended experiential differences. The chapter also discusses implementation constraints and design decisions that affect the comparison.

Chapter 4 presents the study design used to evaluate the two representation modes. It describes the experimental conditions, participant procedure, measures, and analysis approach. Particular attention is given to how embodiment-related constructs, perceived rowing realism, comfort, and preference are measured. The chapter also explains how the comparison is structured to reduce confounds between the two modes.

Chapter 5 reports the results of the evaluation. Quantitative and qualitative findings are presented with respect to the research questions, including measures related to agency, ownership, self-location, embodiment, perceived realism, comfort, and preference where applicable. The chapter focuses on reporting the observed outcomes without extensive interpretation.

Chapter 6 interprets the findings in relation to the theoretical background and research questions. It discusses whether the results support the importance of visuomotor congruence, sport-realistic depiction, or a more nuanced relationship between the two. The chapter also considers limitations of the prototype and study design, as well as implications for future XR sport sys-

tems.

Chapter 7 concludes the thesis by summarizing the main findings and contributions. It revisits the central trade-off between ergometer-congruent and boat-centric representation and outlines directions for future work in VR rowing, embodied interaction, and XR sport design.

Chapter 2

Theoretical Background and Related Work

2.1 Virtual Embodiment

Virtual reality can do more than surround the user with a simulated environment. It can also provide the user with a virtual body, body parts, tools, or task-specific apparatus that appear to occupy the space of action. In such cases, the user does not only perceive a scene from the outside, but may experience a relation between their own body and the body or action represented in the virtual environment. This relation is commonly discussed under the concept of virtual embodiment.

Virtual embodiment is usually not treated as a single, indivisible feeling. Instead, it is often described through multiple related components, especially body ownership, sense of agency, and self-location. Body ownership concerns the experience that a body or body part belongs to oneself. Sense of agency concerns the experience of causing or controlling an action. Self-location concerns where the user experiences themselves to be located in relation to a body or space.

This distinction is important because the components of embodiment can diverge. A user may feel that they control a virtual hand because it moves in response to their own action, while still not strongly feeling that the hand is part of their body. Conversely, a virtual body may appear spatially aligned with the user and therefore support self-location, while delayed or mismatched movement may reduce agency. Embodiment is therefore better understood as a structured experience in which different cues can support different aspects of the relation between the physical and virtual body.

Many embodiment studies build on the logic of body illusion paradigms.

In the rubber hand illusion, for example, synchronous visual and tactile stimulation can lead participants to report ownership over an artificial hand. Virtual hand and virtual body studies extend this logic into immersive environments, where the virtual body can respond to the user’s movement and may be experienced as part of the user’s body representation.

For this thesis, the most relevant aspect of virtual embodiment is not the mere presence of a virtual avatar, but the relation between the user’s real movement and the movement shown in VR. In indoor rowing, the user performs a real, physically effortful action on an ergometer. If a virtual body or rowing apparatus is shown, the system must decide how this physical action is represented. The visible movement may closely correspond to the user’s actual handle and body motion, or it may be transformed into a visually more sport-specific boat and oar action.

This makes VR rowing a useful case for examining embodiment in a tool-mediated and exercise-based context. The user is not only looking at a virtual body, nor only moving a hand in empty space. They are seated on a physical machine, holding a real handle, pushing against foot stretchers, and performing a rhythmic full-body movement. The virtual representation must therefore mediate between the felt physical apparatus and the visual action shown in the virtual environment.

The central concern of this thesis is how this mediation affects embodiment-related experience. An ergometer-congruent representation may support embodiment because the seen action can remain close to the user’s felt movement. A boat-centric representation may support the plausibility and meaning of the sport scenario, but may require a greater transformation between physical action and visual movement. The following sections therefore examine the components of embodiment and the role of visuomotor congruence in more detail.

2.2 Agency, Ownership, and Self-Location

The sense of embodiment is commonly discussed through three partially separable components: sense of agency, body ownership, and self-location. These components are useful for this thesis because the two rowing representations may not affect all aspects of embodiment in the same way. A representation that supports control may not necessarily feel most like rowing, and a representation that looks most like rowing may not necessarily provide the strongest correspondence with the user’s felt body.

Sense of agency refers to the experience of causing or controlling an action. In virtual environments, agency is often supported when a virtual body or

object moves in a way that matches the user’s own intentions and movements. When a user moves their hand and immediately sees a virtual hand perform the same movement, the visual feedback can support the impression that the virtual action is their own. Conversely, delays, spatial offsets, or unexpected transformations can weaken the sense that the observed action is under direct control.

For VR rowing, agency is especially important because the activity is active, continuous, and rhythmic. The user repeatedly initiates and controls a stroke cycle. If the virtual hands, handle, or rowing body visibly follow the user’s real movement, agency may be reinforced throughout the stroke. If the virtual action transforms the user’s movement into a different oar or boat motion, agency may still be present, but it may depend on whether the transformation feels predictable, responsive, and meaningful.

Body ownership refers to the experience that a body or body part is one’s own. Ownership has often been studied through hand and body illusions, where visual, tactile, proprioceptive, and motor cues are combined. Synchronous and spatially plausible cues can support ownership, while incongruent cues can reduce it. In VR, ownership may be influenced by the visual appearance of the avatar, the spatial alignment between the physical and virtual body, and the correspondence between real and virtual movement.

In the context of rowing, body ownership is relevant because the user may see a virtual body performing the rowing action. An ergometer-congruent representation may place the virtual hands and body in closer correspondence with the felt body and handle. This could support ownership because the virtual body appears to occupy a plausible relation to the user’s own body. A boat-centric representation may show a body engaged in a more recognizable rowing action, but if the visual hand or arm movement diverges from the felt handle movement, ownership may be affected differently.

Self-location concerns the experienced location of the self in relation to a body or environment. In immersive VR, the user’s viewpoint is usually placed inside or near a virtual body, which can support the impression of being located in the virtual scene. Self-location may be influenced by visual perspective, body alignment, movement, and multisensory cues.

Self-location is relevant to VR rowing because the user’s physical body is stationary in a room, while the virtual scene may depict movement through water or a rowing environment. The user may feel located on the ergometer, in the virtual boat, or in a hybrid relation between both. In an ergometer-congruent mode, the felt machine and visible apparatus may reinforce the physical setup. In a boat-centric mode, the visual environment may encourage the user to interpret themselves as being in a boat, even though the physical cues still come from the ergometer.

The three components of embodiment therefore provide different lenses on the same mapping problem. Agency emphasizes the relation between intention, movement, and feedback. Ownership emphasizes whether the virtual body or body parts are experienced as belonging to the user. Self-location emphasizes where the user experiences themselves to be situated. A central assumption of this thesis is that the ergometer-congruent and boat-centric representations may produce different patterns across these components rather than a single uniform increase or decrease in embodiment.

2.3 Visuomotor Congruence and Synchrony

Visuomotor congruence describes the correspondence between a user’s physical movement and the movement they see. In virtual embodiment research, this correspondence is one of the central cues linking the physical body to the virtual body. When the user performs an action and the virtual body moves in a spatially and temporally corresponding way, the virtual movement can be interpreted as the visual consequence of the user’s own action.

Two aspects of congruence are especially relevant: temporal synchrony and spatial correspondence. Temporal synchrony concerns whether the virtual movement occurs at the same time as the user’s real movement. Even small delays can affect the perceived connection between action and feedback, particularly for agency. Spatial correspondence concerns whether the virtual movement occurs in the expected position, direction, and configuration relative to the user’s body. A movement can be temporally synchronous but spatially transformed, or spatially aligned but delayed.

The distinction between temporal and spatial congruence matters for VR rowing. A boat-centric representation may respond immediately to the user’s stroke, preserving temporal synchrony, while still transforming the spatial path of the hands into an oar-based movement. An ergometer-congruent representation may preserve both the timing and the approximate spatial logic of the handle pull, because the virtual hands and handle can follow a path closer to the real ergometer movement. Both representations may therefore be responsive, but they differ in how much of the user’s physical movement geometry is preserved.

Classic body illusion research often emphasizes the importance of synchronous multisensory stimulation. In the rubber hand illusion, synchronous visual and tactile cues can support ownership over an artificial hand, while asynchronous cues tend to weaken the illusion. In active virtual hand paradigms, the relationship between the user’s own movement and the seen movement becomes especially important, because movement itself provides a continuous

cue linking the physical and virtual body.

In VR rowing, the user’s hand movement is not isolated from the rest of the body. The stroke is a coordinated sequence involving leg drive, trunk swing, arm pull, handle movement, and seat motion. This makes congruence more complex than simply matching a virtual hand to a real hand. A system can align some cues while transforming others. For example, the timing of the stroke may match the user, while the visible trajectory of the virtual hands may follow an oar-based path rather than the real ergometer handle path. Conversely, an ergometer-congruent representation can align the hand and handle more closely while sacrificing some of the visual realism associated with rowing on water.

This is the core reason why the comparison in this thesis is not a simple test of realism. A realistic-looking boat scene may provide strong contextual plausibility, but still introduce a visuomotor transformation between the felt and seen action. A visually less sport-realistic ergometer representation may provide stronger congruence with the user’s physical action. The two representation strategies therefore emphasize different forms of fidelity.

The expected role of visuomotor congruence is especially strong for sense of agency. If the virtual action closely follows the user’s intended and performed movement, users may experience the action as more directly their own. Body ownership may also benefit from congruence, especially when the virtual hands and body appear spatially aligned with the felt body. Self-location may be affected more indirectly, depending on whether the user experiences themselves as situated in the virtual body, the physical ergometer setup, or the represented rowing environment.

At the same time, congruence should not be treated as the only value in the design of VR sport systems. A perfectly congruent representation of the ergometer may not be the representation users find most meaningful as rowing. A transformed boat-centric representation may reduce spatial congruence while increasing perceived sport realism. This thesis therefore examines visuomotor congruence as a central mechanism, but not as the only determinant of the overall experience.

2.4 Virtual Hands, Tools, and Body Mapping

In many virtual embodiment studies, the virtual hand or body is treated as the primary object of embodiment. For rowing, however, the user’s action is not only body-centred. It is also mediated through a physical handle, a seat, foot stretchers, and, in the boat-centred representation, virtual oars. The user’s movement is therefore distributed across body parts and tools. This

makes the mapping problem more complex than simply placing a virtual hand where the real hand is.

Tools can become closely integrated into action. A rower does not usually experience the handle or oar as a detached object that must be consciously controlled at every moment. Instead, skilled movement often treats the tool as part of the action system. In VR, this matters because the system can choose whether to represent the felt tool directly, as in an ergometer handle, or to transform it into a sport-specific tool, as in a virtual oar.

This distinction is central for the two conditions in this thesis. In the ergometer-congruent mode, the visible apparatus can remain close to the physical one. The user pulls an ergometer handle and sees a virtual handle and body movement that follow the same basic logic. The virtual representation is therefore anchored in the physical apparatus. This may support embodiment because visual, proprioceptive, and haptic cues point toward the same object relation.

In the boat-centred mode, the same physical action is interpreted as rowing a boat. The user still holds a real ergometer handle, but the virtual representation may show hands, oars, blades, and a moving shell. This is more meaningful as a depiction of the sport, but it introduces a representational translation. The felt handle and the seen oar are not identical tools. The system must therefore preserve enough timing and controllability for the user to accept the mapping, while allowing the visual scene to behave like rowing on water.

Virtual hands are especially important in this translation. Hands are visually salient, close to the user's body, and directly involved in the stroke. If virtual hands are spatially aligned with the felt hands, they can strengthen the link between the physical and virtual body. If they are moved into positions required by the oar animation, they may look more correct as rowing hands, but less directly matched to the felt handle path.

The body mapping problem extends beyond the hands. Rowing is a coordinated full-body action. The legs, trunk, arms, and seat movement together define the stroke. A system that tracks only the handle may reproduce the rhythm of rowing but miss important bodily cues. A system that also estimates seat position and body posture can create a more coherent avatar, but still has to decide whether that avatar should follow the ergometer action or the boat action.

For this reason, this thesis treats body mapping as an embodied design variable rather than only an implementation detail. The same physical stroke can be mapped with different priorities. One mapping can privilege correspondence with the user's felt motion. Another can privilege the recognizable structure of on-water rowing. The question is not simply which mapping is

more realistic in a visual sense, but which mapping better supports the user's experience of agency, ownership, and self-location.

2.5 Presence, Vection, and Stationary Self-Motion

Embodiment is related to, but not identical with, presence. Presence concerns the experience of being in a mediated environment, while embodiment concerns the experienced relation between the self and a body or body part. In VR, these experiences often interact. A believable environment may make the virtual body feel more situated, and a coherent body may make the environment feel more actionable. Nevertheless, the concepts should remain separate because a user can feel present in a scene without strongly experiencing a virtual body as their own.

This distinction is important for VR rowing because the boat-centred mode may increase the plausibility of the sport scenario. Seeing water, a shell, oars, and forward movement may make the situation feel more like rowing. This does not automatically mean that embodiment is stronger. A scene can be convincing as a rowing environment while the virtual hands or body still feel less directly connected to the user's own movement.

The visual experience of moving through space is also relevant. In stationary exercise systems, the body remains in one physical location while the virtual environment may depict locomotion. Optic flow can create a visual impression of self-motion, and vection describes the illusion of self-motion in a stationary observer. In rowing, this can make an ergometer action feel connected to movement through a river or race course, even though the user remains in the room.

For this thesis, vection is not the main dependent construct, but it helps explain why a boat-centred representation may feel meaningful. The user's effort produces visible locomotion. The stroke is therefore not only an isolated body movement, but appears to move the user through a virtual environment. This can support plausibility and motivation, and may also influence self-location by encouraging the user to experience themselves as situated in the virtual rowing scene.

At the same time, stationary self-motion is a design tension. Visual motion that is not sufficiently related to the user's action can create a mismatch between visual and bodily cues. In many VR applications, such mismatches are discussed in relation to cybersickness. In rowing, the risk may be moderated by the fact that visible motion is coupled to the user's own rhythmic

effort rather than imposed independently. However, this should be treated cautiously and empirically rather than assumed.

The two rowing representations therefore differ not only in body mapping, but also in how strongly they situate the user in a moving sport scenario. The ergometer-congruent mode may offer stronger correspondence with the physical body and machine. The boat-centred mode may offer stronger environmental plausibility and visually supported self-motion. This makes presence andvection useful background concepts, while the thesis remains focused on embodiment and its subcomponents.

2.6 Rowing Biomechanics and Ergometer Rowing

Rowing is a cyclical, whole-body movement that is commonly described through phases such as the catch, drive, finish, and recovery. During the drive, the rower pushes with the legs, transfers force through the trunk, and completes the stroke with the arms. During the recovery, the sequence is reversed as the rower moves back toward the catch. This rhythm is central to both ergometer rowing and on-water rowing, but the two are not mechanically identical.

An ergometer constrains the rowing action through a fixed machine frame, handle path, sliding seat, and foot stretchers. It allows the user to perform a rowing-like movement indoors and to generate measurable work, but it does not reproduce all forces and spatial relations of a moving boat. In on-water rowing, the rower interacts with oars, blades, riggers, boat movement, and water resistance. The hands and oars follow a rowing-specific action that includes blade handling, squaring and feathering, and the relation between the shell and the water.

This difference is important for the present thesis because the user physically rows on an ergometer while the virtual environment can depict either an ergometer-centred action or an on-water rowing action. A direct one-to-one mapping from physical movement to virtual boat rowing is therefore not possible in a strict biomechanical sense. The system must choose which aspects of the stroke to preserve and which aspects to transform.

The ergometer-congruent representation preserves the indoor rowing apparatus as the main frame of reference. It can align the virtual handle, seat, and body movement with the user's felt setup. This makes the virtual action less like a full on-water rowing scene, but more directly connected to what the user is physically doing.

The boat-centred representation preserves the cultural and sport-specific image of rowing. It can show the user as sitting in a shell, moving through water, and interacting with oars. This may better match what users imagine when they think of rowing as a sport. However, the visual hand and oar movements may need to diverge from the exact physical handle trajectory of the ergometer.

This creates a practical form of fidelity trade-off. Biomechanical or visuomotor fidelity to the ergometer is not the same as contextual fidelity to rowing on water. A system can be faithful to the user’s physical input while looking less like the sport, or faithful to the sport depiction while transforming the user’s physical input. This trade-off provides the concrete movement basis for the embodiment comparison in this thesis.

2.7 VR Sport and Rowing-Related Work

VR sport systems combine bodily action with simulated environments. They can be used for training, exercise, rehabilitation, motivation, or entertainment. Compared with passive VR experiences, sport and exercise systems place greater demands on the relation between movement, feedback, and physical effort. The user is not only observing a scene, but acting continuously within it.

A recurring issue in VR sport is fidelity. Fidelity can refer to the visual realism of the environment, the realism of the task, the correspondence between real and virtual movement, or the usefulness of the simulation for training transfer. These forms of fidelity are related, but they are not the same. A system can look realistic while simplifying movement, or reproduce movement constraints while presenting a more abstract environment.

Rowing has appeared in several virtual or interactive training contexts. Some work has investigated VR or virtual environments for rowing exercise, training, social presence, or transfer between virtual and real rowing. These studies are useful as related work, but the present thesis addresses a more specific embodiment question. It does not primarily evaluate training effectiveness, physiological performance, or long-term motivation. Instead, it asks how two representation strategies affect the experienced relation between the user’s body, movement, and virtual rowing action.

Adjacent exercise systems such as VR cycling are also relevant because they show how stationary physical effort can be connected to virtual locomotion. In such systems, visual motion, environmental context, avatars, and interaction feedback can influence the exercise experience. However, rowing differs from cycling because the hands, handle, seat, trunk, and legs form

a highly visible coordinated movement. The user’s upper body and tool interaction are therefore more central to the embodied visual representation.

The contribution of this thesis is therefore not simply another VR rowing prototype. Its focus is the comparison between an ergometer-centred, high-congruence representation and a boat-centred, sport-depiction representation within the same general VR rowing context. This comparison separates two design values that are often blended together: close correspondence with the user’s actual movement and convincing depiction of the sport scenario.

Related work in VR sport provides the broader design context, while embodiment research provides the conceptual lens. Rowing biomechanics explains why the two mappings are not equivalent. Together, these perspectives motivate the empirical question: when the user rows on an ergometer in VR, is embodiment better supported by preserving the felt movement geometry or by presenting a more recognizable on-water rowing action?

2.8 Synthesis for This Thesis

The theoretical background can be summarized as a mapping problem. Virtual embodiment depends on how the user’s physical body, actions, and sensory feedback are related to a virtual body or action. Agency, ownership, and self-location provide separate but connected ways of describing this relation. Visuomotor congruence, spatial alignment, temporal synchrony, and plausible body mapping are therefore central design variables.

In XR rowing, these variables are unusually concrete. The user performs a real rowing-like movement on a physical ergometer. The system must decide whether the virtual representation should remain close to this physical setup or transform it into a more complete on-water rowing scene. Both strategies are defensible, but they support different forms of realism.

The ergometer-congruent mode prioritizes correspondence with the user’s real action. It is expected to support agency because the visible movement can follow the user’s own stroke closely. It may also support ownership because the virtual hands, body, handle, and seat can be arranged in a spatially plausible relation to the user’s felt body and apparatus. Its limitation is that it may feel less like the culturally recognizable image of rowing on water.

The boat-centred mode prioritizes sport depiction and environmental plausibility. It can show a rower in a shell, moving through water, using oars, and performing a visually recognizable rowing action. This may strengthen presence, self-location in the rowing scene, and the subjective meaning of the activity as rowing. Its limitation is that it may require a transformation between the felt ergometer handle path and the visible oar-based movement.

This thesis therefore does not assume that one representation is universally better. Instead, it expects that the two modes may produce different embodiment profiles. The ergometer-congruent mode may be stronger for direct agency and body correspondence. The boat-centred mode may be stronger for plausibility, sport realism, and the feeling of being situated in a rowing environment. Body ownership and self-location may depend on how participants integrate these cues.

The empirical comparison is built around this tension. By holding the general VR rowing context constant while varying the representation strategy, the study examines whether embodiment in XR rowing is more strongly supported by precise movement correspondence or by a more realistic sport depiction. The result should help clarify how embodiment concepts can inform the design of future XR exercise and sport systems.

Chapter 3

System and Prototype

Chapter 4

Study Design and Method

4.1 Study Design

The study evaluates the XR rowing prototype by comparing two representation modes for the same physical rowing action. In both conditions, participants row on the same ergometer setup. What changes is not the exercise task itself, but the representational relation between the felt ergometer movement and the visible rowing body, handle, oars, and environment.

The first condition is an ergometer-congruent representation. It is designed to preserve close spatial and temporal correspondence between the participant's real interaction with the ergometer and the virtual rowing body. The visible hands, handle relation, seat movement, and body movement are therefore intended to remain as close as possible to the physical setup. This condition prioritizes embodied action congruence: the participant should see a rowing body whose movement appears directly tied to the movement they perform on the machine.

The second condition is a boat-realistic representation. It is designed to make the action legible as rowing on water by showing a rowing body in a boat-like context with virtual oars. This representation is motivated by the fact that ergometer rowing can approximate important parts of rowing movement, but does not reproduce all on-water rowing mechanics, especially oar- and blade-related actions (Lamb, 1989; Soper & Hume, 2004). The boat-realistic condition therefore prioritizes sport-action depiction even where the virtual oar path cannot be a literal copy of the real ergometer handle path.

The planned design is within-subjects: every participant will experience both modes. This design is appropriate because the comparison concerns two alternative representations of one physical action, allowing each participant to serve as their own control. The condition order will be counterbalanced

where possible to reduce order, learning, adaptation, and fatigue effects.

TODO: TODO:CONFIRM final counterbalancing scheme, condition labels, and whether the order assignment is random, alternating, or manually balanced.

The first data-collection version does not include physiological threat measures, heart-rate measures, or a hand-offset probe. This keeps the initial evaluation focused on the main representation comparison: whether the two modes differ in reported ownership, agency, embodied action congruence, virtual hand/oar mapping, sport-action plausibility, perceived mismatch, and preference. Comfort and sickness symptoms will be handled as safety and debriefing observations rather than as embodiment outcomes.

4.2 Participants and Recruitment

Participants will be recruited for a short XR rowing evaluation session. Because the study evaluates an early prototype and a specific representation trade-off, recruitment will be treated as exploratory rather than as a claim to represent all rowers, all VR users, or all exercise users. The final participant number, recruitment route, inclusion criteria, and exclusion criteria must be fixed before data collection is reported in past tense.

TODO: TODO:CONFIRM participant count, recruitment source, age range, inclusion criteria, exclusion criteria, and whether formal ethics approval or supervisor approval is required.

The study materials will record background variables that may influence how participants interpret the two representations. These include prior VR experience, prior rowing or ergometer experience, susceptibility to VR sickness or motion sickness, and any health or comfort restrictions relevant to rowing with a head-mounted display. These variables are not used to classify participants into experimental groups in the first study version, but they provide context for interpreting questionnaire responses and open comments.

Participants will receive the same explanation of the task and will be told that the study concerns their subjective experience of two virtual rowing representations. The instructions should avoid suggesting that one mode is expected to be better, more correct, or more realistic overall. This is important because the thesis does not treat ergometer congruence and boat realism as a simple hierarchy; it treats them as competing design priorities.

4.3 Apparatus and Study Setting

The apparatus consists of the physical rowing ergometer, the VR headset, the tracking and calibration setup, and the Unity-based rowing prototype described in Chapter 3. The same physical rowing task and the same basic prototype environment will be used in both conditions. This reduces the risk that differences in ratings are caused by changes in exercise task, exertion context, or study setting rather than by the representation mode.

TODO: TODO:CONFIRM exact ergometer model, headset, tracking devices, computer or standalone setup, software version, and room setup.

The participant interacts with the real ergometer while wearing the headset. The handle relation is tracked through the participant's hands, while the remaining rowing-body motion is inferred from the aligned rowing machine, calibrated setup geometry, and participant body measurements. This makes calibration part of the experimental apparatus rather than a neutral preparation step: if the machine, body scale, or virtual rowing frame are poorly aligned, the resulting ratings may reflect setup error rather than the intended contrast between ergometer-congruent and boat-realistic representation.

TODO: TODO:CONFIRM exact hand tracking method, body-measurement inputs, calibration steps, approximate calibration duration, and any measured or estimated end-to-end latency.

Latency and responsiveness will be documented where possible because delayed closed-loop avatar feedback can influence agency, ownership, simultaneity judgments, and motor performance (Waltemate et al., 2016).

4.4 Procedure

Each session will begin with participant information, consent, and a short health and comfort check. The participant will then be introduced to the rowing setup and the VR headset in neutral terms. The experimenter will explain that the study compares two virtual representations of the same rowing action and that the participant should report their own experience rather than trying to identify a correct or preferred technical solution.

Before the measured trials, the system will be calibrated to the participant and the physical rowing setup. The participant will then complete both representation conditions. In each condition, the participant will first receive a short familiarization period, followed by a standardized rowing period. The familiarization period is included because movement-based ownership can take time to emerge, and very short exposures may capture initial confusion rather than a stabilized experience of the mapping (Kalckert & Ehrsson,

2017).

TODO: TODO:CONFIRM exact familiarization duration and exact rowing duration per condition. Current default: at least one minute of stable familiarization followed by a brief standardized rowing period.

After each condition, the participant will complete the per-condition questionnaire. The questionnaire will include ownership and agency items, exploratory body-location items, and thesis-specific items on embodied action congruence, the real handle as a physical anchor, virtual hand/oar mapping, sport-action plausibility, and perceived mapping mismatch. Safety and comfort symptoms will be recorded separately from these construct ratings.

A short rest will be offered between conditions. After both conditions have been completed, the participant will answer comparative preference questions and may provide open comments about the main differences they noticed. The participant may stop the session at any time, and the experimenter will stop the session if the participant reports nausea, dizziness, pain, distress, or unsafe discomfort.

4.5 Embodiment Measures

The primary embodiment outcomes are sense of ownership and sense of agency. These constructs match the central comparison of the study: the ergometer-congruent mode is designed to make the visible rowing body closely follow the participant’s actual movement, while the boat-realistic mode may transform the visible hand and oar motion in order to depict rowing on water. The study therefore asks not only which representation participants prefer, but whether the virtual rowing body and movement feel like their own body and action under each mapping strategy.

The planned primary questionnaire basis is the Virtual Embodiment Questionnaire (VEQ), focusing on the Ownership/Acceptance and Control/Agency subscales (Roth & Latoschik, 2020). These subscales will be administered after each condition and scored separately. The VEQ Change subscale is not included as a primary outcome because the study does not primarily manipulate avatar body shape, body size, or body-schema change.

TODO: TODO:VERIFY final VEQ item wording against the original questionnaire sheet before data collection.

Because self-location is conceptually relevant but is not preserved as a final VEQ factor, the study includes exploratory body-location items adapted from avatar embodiment questionnaire practice (Gonzalez-Franco & Peck, 2018). These items are analyzed separately from the VEQ subscales. They are included because the two representation modes may differ not only in

control and ownership, but also in whether the participant feels spatially aligned with the visible rowing body and hands.

Embodiment will not be treated as a single undifferentiated score. Ownership, agency, and exploratory body-location ratings will be reported separately so that the two representation modes can produce different profiles. For example, a representation could feel more directly controllable without necessarily being preferred overall, or it could look more like on-water rowing while still producing a stronger sense of handle/hand mismatch.

4.6 Rowing-Specific and Experience Measures

The established embodiment items are complemented by custom rowing-specific items. These items are necessary because the thesis does not compare embodiment in an abstract avatar task. It compares two ways of representing the same real ergometer action: one that preserves direct correspondence with the physical machine, and one that restores a more recognizable boat-and-oar rowing depiction. Rowing biomechanics and simulator work make this distinction meaningful rather than cosmetic: a central ergometer handle can support controlled indoor rowing, but it does not reproduce all on-water oar handling, blade, and boat mechanics (Lamb, 1989; Rauter et al., 2013; Soper & Hume, 2004).

The custom per-condition items therefore target thesis-specific dimensions that the VEQ does not measure directly:

- embodied action congruence between the participant’s real ergometer movement and the virtual rowing body,
- the real handle as a physical, haptic, and proprioceptive anchor for the seen virtual hands, handle, or oars,
- acceptability of the virtual hand/oar mapping, especially where the boat-realistic mode retargets the real handle relation into an oar-like action,
- sport-action plausibility of the rowing depiction, including whether the shown body, oars, and stroke phases read as rowing on water rather than generic seated movement,
- perceived conflict between the felt real handle path and the seen virtual hand, handle, or oar path.

These custom items are exploratory thesis-specific measures. They will not be mixed into VEQ Ownership or VEQ Agency scores. Negatively phrased mismatch items will be reported separately unless a final scoring scheme is explicitly defined. This separation is important because a participant may rate the boat-realistic mode as more recognizable rowing while also reporting weaker direct control or stronger handle/oar mismatch.

Comfort, dizziness, nausea, pain, or unsafe discomfort will be recorded through safety checks and debriefing notes. These observations are relevant for feasibility and limitations, but they are not interpreted as embodiment, agency, or sport-action plausibility scores.

After both conditions, participants will answer comparative preference questions. These will ask which representation they preferred overall, which felt more directly connected to the real ergometer movement, and which looked more like rowing on water. Open comments will be collected to identify mismatch sources and details that are not captured by numeric ratings. **TODO: TODO:CONFIRM final custom item list and response scale before pilot testing.**

4.7 Data Handling and Scoring

Questionnaire responses will be recorded per participant and per condition. The analysis table will include a participant code, condition order, background variables, per-condition ratings, comparative preference answers, and optional open comments. Condition order must be retained because the within-subject design makes order, learning, and fatigue possible interpretive factors.

Participant names or direct identifiers should not be stored in the analysis table. If contact or consent information must be stored, it should be kept separately from questionnaire responses.

The VEQ Ownership/Acceptance score will be calculated as the mean of the selected ownership items. The VEQ Control/Agency score will be calculated as the mean of the selected agency items. If any VEQ item is adapted for wording, the adaptation must be documented in Appendix A. Exploratory body-location items will be reported separately because they are not part of the final VEQ factor structure used as the primary measure.

Custom rowing-specific items will be scored and reported separately from the VEQ. Preference items will be analyzed after both conditions because they depend on direct comparison. Open comments will be used to contextualize questionnaire patterns, especially where participants describe concrete sources of congruence, mismatch, comfort, or confusion.

TODO: TODO:CONFIRM file format, anonymization process, missing-data handling, and whether open comments are translated, paraphrased, or quoted.

4.8 Analysis Plan

The analysis will compare the ergometer-congruent and boat-realistic conditions within participants. The primary comparisons are VEQ Ownership/Acceptance and VEQ Control/Agency. These outcomes directly address whether either representation better supports the subjective feeling that the virtual rowing body belongs to the participant and that its movements are caused and controlled by the participant’s own rowing action.

Exploratory comparisons will include body-location items, embodied action congruence, physical handle anchoring, virtual hand/oar mapping acceptability, sport-action plausibility, mapping mismatch reports, and preference. These outcomes are not treated as validated embodiment subscales. Instead, they are used to interpret how the two representation strategies differ as rowing interfaces.

Descriptive statistics will be reported for each condition and measure. These will include at minimum the mean, median, standard deviation or interquartile range, and the within-participant difference where appropriate. Visual summaries may be used to show the direction and consistency of within-participant differences.

If the final sample size and response distributions support it, paired statistical tests will be used for the main within-subject comparisons. A paired t-test may be used for approximately symmetric interval-style scale differences; a Wilcoxon signed-rank test may be used as a non-parametric fallback. The chosen test must be stated before final reporting.

TODO: TODO:CONFIRM final statistical tests, alpha level, effect-size reporting, and whether the analysis is primarily descriptive due to sample size.

The interpretation will remain cautious. Higher ratings in one mode will be described as participant-reported differences in this prototype and study setup, not as proof that one representation is universally superior. The analysis is expected to be most informative when the measures diverge, for example if one condition is preferred or judged more rowing-like while the other is rated as more directly controllable.

4.9 Ethics, Safety, and Method Limitations

The study involves physical exertion on an ergometer while wearing a VR headset. The procedure must therefore include basic safety screening, clear stop criteria, and the option to withdraw at any time without giving a reason. These safety procedures are part of the method because fatigue, discomfort, or headset-related symptoms could affect both participant wellbeing and the interpretation of ratings.

Participants will be informed that they may experience physical fatigue, discomfort, dizziness, or VR-related sickness symptoms. They will be instructed to stop immediately if they feel unsafe or unwell. The experimenter will monitor the session and pause or terminate the trial if necessary.

TODO: TODO:CONFIRM consent form, data protection wording, local ethics or supervisor approval requirements, and final safety checklist.

The first study version deliberately avoids threat stimuli, physiological response measures, and hand-offset probes. This reduces additional ethical complexity and keeps the first evaluation focused on the main representation comparison. If these probes are added in a later study version, they should be described as separate measures with separate justification rather than folded into the current questionnaire design.

The method has expected limitations. It relies mainly on self-report, uses a prototype-specific implementation, and may involve a small or convenience-based sample. The results should therefore be interpreted as evidence about the studied XR rowing setup rather than as general conclusions about all VR rowing systems or all embodiment mechanisms. The final discussion should also distinguish between limitations of the study design and limitations of the prototype implementation, because a low rating may reflect the representation concept, the current calibration/tracking quality, or both.

Chapter 5

Results

Chapter 6

Discussion

Chapter 7

Conclusion

Appendix A

Questionnaires

A.1 Questionnaire Overview

This appendix records the planned questionnaire materials for the XR rowing study. The final participant-facing wording must be verified before data collection.

TODO: TODO:VERIFY final English wording against the original questionnaire sources.

TODO: TODO:VERIFY-DE final German wording with a human speaker before use.

TODO: TODO:CONFIRM final item order, response scale labels, and whether items are randomized or grouped by construct.

Unless changed before data collection, per-condition ratings will use a 7-point agreement scale:

1 = strongly disagree, 7 = strongly agree

The VEQ ownership and agency items are the primary embodiment measures. Rowing-specific items are custom exploratory measures and must not be combined with VEQ scores.

The custom items are intended to separate embodied action congruence from sport-action depiction. They therefore ask about the real ergometer action, the real handle as a physical anchor, virtual hand/oar mapping, rowing-on-water plausibility, and perceived mismatch between the felt handle movement and the seen virtual movement. These constructs are thesis-specific and should be reported separately from the validated VEQ subscales.

Safety and comfort checks are recorded separately from the questionnaire constructs. Reports of dizziness, nausea, pain, or unsafe discomfort are relevant for stopping rules, feasibility, and limitations, but they should not be combined with VEQ or rowing-specific exploratory scores.

A.2 Per-Condition Questionnaire: English Draft

Participants complete this questionnaire after each rowing condition.

Embodiment: Ownership

TODO: TODO:VERIFY these VEQ items against the original VEQ questionnaire sheet before data collection.

1. It felt like the virtual body was my body.
2. It felt like the virtual body parts were my body parts.
3. The virtual body felt like a human body.
4. It felt like the virtual body belonged to me.

Embodiment: Agency

1. The movements of the virtual body felt like they were my movements.
2. I felt like I was controlling the movements of the virtual body.
3. I felt like I was causing the movements of the virtual body.
4. The movements of the virtual body were in sync with my own movements.

Exploratory Body Location

1. I felt as if my body was located where I saw the virtual rowing body.
2. I felt as if my hands were located where I saw the virtual hands.

Rowing-Specific Exploratory Items

1. The virtual rowing body felt connected to the movement I performed on the real ergometer.
2. The real handle I was holding fit with the virtual hands, handle, or oars I saw.
3. The real handle I was holding made the virtual rowing action feel physically grounded.

4. The virtual hand and arm movement made sense in relation to my real handle movement.
5. The virtual oar or rowing motion felt like a controllable extension of my own action.
6. The representation felt believable as rowing on water.
7. The visible hand, handle, or oar path made sense for the rowing representation I saw.
8. I noticed a conflict between the real handle movement and the virtual hand or oar movement.

Safety and Comfort Check

Did this condition make you feel dizzy, nauseous, in pain, or unsafe? If yes, please tell the experimenter and briefly describe it.

Open Comment

What, if anything, felt especially convincing or mismatched about the relation between the real ergometer and the virtual rowing representation?

A.3 Per-Condition Questionnaire: German Draft

Teilnehmende füllen diesen Fragebogen nach jeder Ruderbedingung aus.

TODO: TODO:VERIFY-DE German wording before participant use. This draft uses ASCII transliterations and is not final.

Verkörperung: Körperzugehörigkeit

1. Es fühlte sich so an, als wäre der virtuelle Körper mein Körper.
2. Es fühlte sich so an, als wären die virtuellen Körperteile meine Körperteile.
3. Der virtuelle Körper fühlte sich wie ein menschlicher Körper an.
4. Es fühlte sich so an, als gehöre der virtuelle Körper zu mir.

Verkoerperung: Handlungsgefuehl

1. Die Bewegungen des virtuellen Koerpers fuehlten sich wie meine Bewegungen an.
2. Ich hatte das Gefuehl, die Bewegungen des virtuellen Koerpers zu kontrollieren.
3. Ich hatte das Gefuehl, die Bewegungen des virtuellen Koerpers zu verursachen.
4. Die Bewegungen des virtuellen Koerpers waren mit meinen eigenen Bewegungen synchron.

Explorativ: Koerperort

1. Ich hatte das Gefuehl, dass mein Koerper dort war, wo ich den virtuellen Ruderkoerper sah.
2. Ich hatte das Gefuehl, dass meine Haende dort waren, wo ich die virtuellen Haende sah.

Ruderspezifische explorative Items

1. Der virtuelle Ruderkoerper fuehlte sich mit meiner Bewegung auf dem echten Ergometer verbunden an.
2. Der echte Griff in meinen Haenden passte zu den virtuellen Haenden, dem virtuellen Griff oder den virtuellen Rudern, die ich gesehen habe.
3. Der echte Griff in meinen Haenden gab der virtuellen Ruderbewegung eine plausible koerperliche Grundlage.
4. Die virtuellen Hand- und Armbewegungen ergaben im Verhaeltnis zu meiner echten Griffbewegung Sinn.
5. Die virtuelle Ruder- oder Bootsbewegung fuehlte sich wie eine kontrollierbare Erweiterung meiner eigenen Handlung an.
6. Die Darstellung fuehlte sich wie glaubwuerdiges Rudern auf dem Wasser an.
7. Die sichtbare Hand-, Griff- oder Ruderbahn passte zu der Ruderdarstellung, die ich gesehen habe.

8. Ich habe einen Konflikt zwischen der echten Griffbewegung und der virtuellen Hand- oder Ruderbewegung bemerkt.

Sicherheits- und Komfortcheck

Hat diese Bedingung Schwindel, Uebelkeit, Schmerzen oder ein unsicheres Gefuehl ausgeloeost? Falls ja, sagen Sie bitte der Versuchsleitung Bescheid und beschreiben Sie es kurz.

Freier Kommentar

Was fuehlte sich an der Beziehung zwischen echtem Ergometer und virtueller Ruderdarstellung besonders ueberzeugend oder unpassend an?

A.4 Post-Comparison Questionnaire: English Draft

Participants complete these questions after both rowing conditions.

1. Which representation did you prefer overall?
2. Which representation felt more directly connected to your real ergometer movement?
3. Which representation looked more like rowing on water?
4. In which representation did the relation between the real handle and the virtual hands or oars feel more convincing?
5. Which representation would you rather use again?

Safety and Comfort Follow-Up

Did either representation feel less comfortable or cause dizziness, nausea, pain, or unsafe discomfort? If yes, which one and what did you notice?

Open Comparison Comment

Please describe the main difference you noticed between the two rowing representations.

TODO: TODO:CONFIRM whether preference questions use forced choice, 7-point comparison scales, or both.

A.5 Post-Comparison Questionnaire: German Draft

Teilnehmende füllen diese Fragen nach beiden Ruderbedingungen aus.

TODO: TODO:VERIFY-DE German wording before participant use. This draft uses ASCII transliterations and is not final.

1. Welche Darstellung haben Sie insgesamt bevorzugt?
2. Welche Darstellung fühlte sich direkter mit Ihrer echten Ergometerbewegung verbunden an?
3. Welche Darstellung sah stärker nach Rudern auf dem Wasser aus?
4. Bei welcher Darstellung fühlte sich die Beziehung zwischen echtem Griff und virtuellen Händen oder Rudern überzeugender an?
5. Welche Darstellung würden Sie lieber noch einmal verwenden?

Nachfrage zu Sicherheit und Komfort

Fühlte sich eine der Darstellungen weniger angenehm an oder hat sie Schwindel, Übelkeit, Schmerzen oder ein unsicheres Gefühl ausgelöst? Falls ja, welche Darstellung war es und was haben Sie bemerkt?

Freier Vergleichskommentar

Bitte beschreiben Sie den wichtigsten Unterschied, den Sie zwischen den beiden Ruderdarstellungen bemerkt haben.

Appendix B

Study Materials

B.1 Study Materials Overview

This appendix records participant-facing study materials and experimenter scripts. The final versions must be checked before data collection.

TODO: TODO:CONFIRM consent process, data protection wording, and local ethics or supervisor requirements.

TODO: TODO:VERIFY-DE German participant-facing wording before use.

The materials are intended to support a short within-subject XR rowing session in which each participant tries both representation modes and answers questionnaires after each mode.

B.2 Participant Information: English Draft

You are invited to take part in a short study about virtual reality rowing. In the study, you will row on a physical rowing ergometer while wearing a VR headset. You will try two different virtual rowing representations and answer short questionnaires about how each representation felt.

The study focuses on how the virtual rowing body and movement are experienced. There are no right or wrong answers. Please answer based on your own experience.

Participation is voluntary. You may stop at any time without giving a reason. Please tell the experimenter immediately if you feel dizzy, nauseous, uncomfortable, in pain, or unsafe.

The session may involve physical exertion. Please do not participate if rowing, VR headset use, or moderate physical exertion would be unsafe for you.

TODO: TODO:CONFIRM exact duration, data storage statement, anonymity or pseudonymity wording, and consent wording.

B.3 Participant Information: German Draft

TODO: TODO:VERIFY-DE German wording before participant use. This draft uses ASCII transliterations and is not final.

Sie sind eingeladen, an einer kurzen Studie zu virtuellem Rudern teilzunehmen. In der Studie rudern Sie auf einem echten Ruderergometer, waehrend Sie ein VR-Headset tragen. Sie probieren zwei unterschiedliche virtuelle Ruderdarstellungen aus und beantworten kurze Frageboegen dazu, wie sich jede Darstellung angefuehlt hat.

Die Studie untersucht, wie der virtuelle Ruderkoerper und die virtuelle Bewegung erlebt werden. Es gibt keine richtigen oder falschen Antworten. Bitte antworten Sie nach Ihrem eigenen Eindruck.

Die Teilnahme ist freiwillig. Sie koennen jederzeit ohne Angabe von Gruenden abbrechen. Bitte sagen Sie der Versuchsleitung sofort Bescheid, wenn Ihnen schwindlig oder uebel wird, wenn Sie Schmerzen haben, sich unwohl fuehlen oder sich unsicher fuehlen.

Die Sitzung kann koerperlich anstrengend sein. Bitte nehmen Sie nicht teil, wenn Rudern, die Nutzung eines VR-Headsets oder moderate koerperliche Anstrengung fuer Sie unsicher waere.

TODO: TODO:CONFIRM exact duration, data storage statement, anonymity or pseudonymity wording, and consent wording.

B.4 Session Script: English Draft

Before Starting

1. Welcome the participant.
2. Explain the task in neutral terms.
3. Confirm consent and safety screening.
4. Explain that the participant may stop at any time.
5. Fit the VR headset and check comfort.
6. Calibrate the rowing setup.

Condition Procedure

1. Start the assigned first condition.
2. Allow the participant to familiarize themselves with the mapping.
3. Begin the standardized rowing period.
4. Stop the condition and remove or pause the headset as needed.
5. Ask the participant to complete the per-condition questionnaire.
6. Offer a short rest.
7. Repeat the procedure for the second condition.

After Both Conditions

1. Ask the participant to complete the post-comparison questionnaire.
2. Ask whether they want to add any comments.
3. Thank the participant and debrief them.

TODO: TODO:CONFIRM exact wording for condition names, familiarization duration, rowing duration, rest duration, and debrief.

B.5 Session Script: German Draft

TODO: TODO:VERIFY-DE German wording before participant use. This draft uses ASCII transliterations and is not final.

Vor dem Start

1. Teilnehmende Person begrüessen.
2. Aufgabe neutral erklären.
3. Einwilligung und Sicherheitscheck bestätigen.
4. Erklären, dass die Teilnahme jederzeit beendet werden kann.
5. VR-Headset anpassen und Komfort prüfen.
6. Rudersystem kalibrieren.

Ablauf pro Bedingung

1. Erste zugewiesene Bedingung starten.
2. Eingewoehnung in die Bewegungszuordnung erlauben.
3. Standardisierte Ruderphase beginnen.
4. Bedingung beenden und Headset bei Bedarf absetzen oder pausieren.
5. Fragebogen zur Bedingung ausfuellen lassen.
6. Kurze Pause anbieten.
7. Ablauf fuer die zweite Bedingung wiederholen.

Nach beiden Bedingungen

1. Vergleichsfragebogen ausfuellen lassen.
2. Nach freiwilligen Zusatzkommentaren fragen.
3. Teilnehmende Person bedanken und debriefen.

TODO: TODO:CONFIRM exact wording for condition names, familiarization duration, rowing duration, rest duration, and debrief.

B.6 Safety Checklist

Before Participation

- Confirm that the participant understands the task.
- Confirm that participation is voluntary.
- Ask about acute injuries or pain relevant to rowing.
- Ask about dizziness, strong VR sickness susceptibility, or motion sickness concerns.
- Ask whether moderate physical exertion is safe for the participant.
- Check that the rowing area is clear and the headset fit is stable.

Stop Criteria

Stop or pause the session if the participant reports:

- dizziness,
- nausea,
- pain,
- distress,
- unsafe discomfort,
- headset or ergometer fit problems,
- a request to stop.

TODO: TODO:CONFIRM whether any formal exclusion criteria or supervisor-approved safety wording must be added.

Appendix C

Additional Results

Appendix D

System Details

Appendix E

AI Assistance Statement

This appendix documents how AI tools (e.g., ChatGPT, Codex) were used in the research and writing of this thesis. It summarizes the nature of the assistance, how the outputs were verified, and the role of the author in reviewing and editing AI-generated content.

The repository includes an `ai_logs/` folder for process notes about the use of AI and large language model tools during thesis work. These logs are intended to support transparency about drafting, restructuring, review, LaTeX maintenance, and citation-gap detection. They are not treated as thesis evidence or as a substitute for source verification.

The author remains responsible for the final text, scientific claims, study design, data collection, analysis, interpretation, and citations.

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